MANIPAL INSTITUTE OF TECHNOLOGY

Manipal Academy of Higher Education, Manipal

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

THIRD SEMESTER B.TECH. END SEMESTER EXAMINATION - 2025

CS 2102: DIGITAL SYSTEMS AND COMPUTER ORGANIZATION

Branch: Computer Science & Engineering (CSE) | Credit: 4

Time: 3 Hours | Max. Marks: 50

Instructions: Answer ALL questions. Each question carries designated marks.

PART A

1. (a) Design a synchronous mod-12 counter using JK flip-flops and derive state equations. [5 Marks]

1. (b) Explain Booth multiplication algorithm with signed integers 11 x -5 step-by-step. [5 Marks]

PART B

2. (a) Analyze 5-stage RISC instruction pipelining and mitigation of load-use data hazards. [5 Marks]

2. (b) Differentiate Direct Mapping, Fully Associative, and Set Associative Cache organizations. [5 Marks]

PART C

3. (a) Explain virtual memory management: Page Tables, TLB hit/miss penalties, and LRU replacement. [10 Marks]

PART D

4. (a) Describe DMA controller cycle stealing mode vs burst mode in I/O subsystem. [10 Marks]

[END OF QUESTION PAPER - MIT MANIPAL CSE 3RD SEMESTER ARCHIVE]